



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

MBA-Tech/

Program: B Tech Stream ;Data Science

Year: II Semester: IV

Subject: Database Management Systems

Time: 3 hrs (10:00am to 1:00pm)

Date: 27 /Apr/ 23

No. of Pages: 1 to 3

Marks:100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	[20]
CO-1 ; SO-1 ; BL-2	a.	Explain specialization and generalization with suitable examples.	[5]
CO-4 ; SO- 1; BL-2	b.	Write a short note on conflict and view serializability.	[5]
CO- 2; SO-1 ; BL-2	c.	List the aggregate functions supported by SQL?	[5]
CO- 4; SO-1 ; BL-2	d.	Write down the feature of No SQL	[5]

Q2 CO-1; SO-1; BL-1	a.	Consider an airline database. The database must keep track of customers and their reservations, flights and their status, seat assignments on individual flights, and the schedule and routing of future flights. Construct an E-R diagram for the airline database. List down all assumptions that you make about the mapping constraints.	[10]
CO-1; SO-2; BL-2	b.	Define Data models and its types in detail.	[10]
Q3 CO-3; SO-1; BL-3	a.	Consider a relation R(ABCDE) with FDs {AB → C, B → D, D → E } i. Identify the candidate keys ii. Identify the current Normal form of the Relation iii. Decompose the relation and normalize it to 3NF	[10]
CO-3; SO-1; BL-3	b.	Discuss insertion, deletion, and modification anomalies. Why are they considered bad? Illustrate with examples.	[10]
Q4 CO-2; SO-1; BL-1	a.	Explain the distinctions among the terms primary key, candidate key, and superkey by taking suitable examples.	[10]
CO-1; SO-1; BL-3	b.	Explain Three Tier architecture with suitable diagrams. Why do we need mappings between schema levels	[10]
Q5 CO-2; SO-2; BL-3	a.	Consider the following relations schemas: Employee(SSN, Name, Salary, Dept No, MGRSSN) Customer(AccountNo, Name, LenderSSN) Loan(LoanNumber, Amount, AccountNo) Specify the following queries in SQL: i. Write down the primary key and foreign key of all tables in the database.	[10]

		ii. Find the numbers (Dept No) of all departments where the average salary of an employee in that department is strictly greater than the average salary of employees in the company. iii. Find the SSN of all employees whose managers do not have any customers iv. Find the name of the employee with the maximum salary. v. For each employee, retrieve the employee's name, and the name of his or her immediate manager name.	
CO-2; SO-1; BL-2	b	List the operations of relational algebra and the purpose of each.	[10]
Q6 CO-2; SO-1; BL-2	a.	Explain the types of joins with examples.	[10]
CO-4; SO-1; BL-2	b.	Explain the data flow between various states of transaction. and List down the ACID properties. Explain the usefulness of each.	[10]
Q7 CO-4; SO-1; BL-2	a	Explain the storage types of NoSQL in detail.	[10]
CO-4; SO-1; BL-2	b	Why do we need concurrency control? Explain it. Illustrate Concurrent execution of transactions with examples?	[10]